

Srishti Lamba

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Education

Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT) Gandhinagar, Gujarat
M.Sc. Data Science | CPI: 8.66 2025 - Present

Maharaja Sayajirao University of Baroda Vadodara, Gujarat
B.Sc. Statistics (Honours) | CPI: 6.57 2021 - 2024

Projects

- 🔗 [Live App] **RupeeRisk: USD/INR Forecasting** | Python, BigQuery, Scikit-learn, Statsmodels, Streamlit
 - Engineered a **four-channel geopolitical tension index** from GDELT Goldstein Scale data via Google BigQuery, and lagged features by one week to eliminate look-ahead bias.
 - Benchmarked **9 forecasting models** (ARIMA, ARIMAX, Lasso, Random Forest, Gradient Boosting, and hybrids) across a **200-week rolling backtest**; ARIMAX led on risk-adjusted return (**1.30 Sharpe**) and Lasso on directional accuracy (**58.5% MDA**), with USD/INR stationarity confirmed ($ADF = -7.64, p < 0.0001$).
 - Evaluated strategy returns (**17.67% cumulative**) against India's 6.5% risk-free hurdle and deployed an interactive **Streamlit dashboard** with real-time pipeline controls and next-week signal generation.
- 🔗 **Digital Consumption & Academic Performance** | Python, Statistical Modeling, Longitudinal Study Design Ongoing
 - Co-designed a **multi-wave longitudinal study** (pilot N=9; main N=32) combining objective phone-tracking (StayFree) with cognitive tests (SART, PVT-B, Dot-Probe) and validated psychometric scales (BSMAS, PHQ-4, CFS, AMS, KSS), under **Prof. Bhaskar Chaudhury**, DA-IICT.
 - Tested **26 pre-registered hypotheses**; identified a significant **8.32 hrs/day** self-reported screen-time underestimation ($p = 0.0015$), revealing a systematic self-report bias with implications for survey-based measurement.
 - Confirmed path regression models linking social-media use to cognitive fatigue ($p = 0.0008$) and focus impairment ($p = 0.0178$); extending to a 3-wave longitudinal panel with Cross-Lagged Panel Models and LOOCV risk scoring.
- 🔗 [Live App] [Tableau] **IndiaShield: Defense Budget Intelligence Platform** | Python, Tableau, Streamlit, Pandas
 - Built a **6-module strategic intelligence platform** consolidating SIPRI military expenditure, Union Budget, and MoD Annual Report data with real NSE market data across **24 years**.
 - Surfaced a "Modernisation Paradox" where capital acquisition share **fell from 38.3% to 28.9%** (FY16-FY25) despite $2.5\times$ nominal budget growth.
 - Published a parallel **Tableau Public** dashboard with custom Sankey diagrams, choropleths, and treemaps, paired with an interactive Streamlit app for dynamic filtering.
- 🔗 [Live App] **LogiSense: Logistics Analytics Dashboard** | PostgreSQL, Python, Netlify
 - Designed and normalized a **7-table BCNF PostgreSQL** schema for a logistics operations database as part of DS-604 (DBMS) coursework, collaborating on a 3-person team.
 - Implemented adaptive **IQR-based anomaly detection** to flag shipment and delivery irregularities across the data pipeline.
 - Iterated the platform across multiple build cycles before deploying the final interactive dashboard to **Netlify** for filtering and visualizing shipment metrics.

Technical Skills

Programming Languages: Python, R, SQL, Microsoft Excel

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, Statsmodels

Databases: PostgreSQL, Google BigQuery, SQLite

Data Visualization: Tableau, Power BI, Plotly, Matplotlib, Streamlit

Statistics & Modeling: Time-Series Forecasting, Regression Analysis, Hypothesis Testing, ADF Stationarity Testing, Longitudinal Study Design

Developer & Cloud Tools: Git, Jupyter Notebook, Streamlit Cloud, Netlify

Certifications & Achievements

- **Google Analytics 4 (GA4) Certification** | Google
- **Product Analytics Certification** | Pandora
- Secured **3rd Place** at Regional-Level Science Exhibition for presenting a Green Hydrogen Smart Village model.